The synchronized companion on an interior record-renewal tail

27 August 2026

Let

$$x_{t+1} = \frac{3x_t + 1}{2^{a_t}}, \qquad A_t := \sum_{h<t} a_h, \qquad q_t := \frac{3^t}{2^{A_t}} \tag{1}$$

be a normalized infinite accelerated orbit, with $q_t \to \infty$. Its fixed synchronized companion is the positive real sequence satisfying

$$2^{a_t} e_{t+1} = 3e_t - 1, \qquad e_t > 1, \qquad \frac{e_t}{q_t} \longrightarrow 0. \tag{2}$$

Equivalently,

$$e_t = \frac{q_t}{3} \sum_{h=t}^{\infty} \frac{1}{q_h}. \tag{3}$$

Let $j_0 < j_1 < \cdots$ be consecutive coefficient tail records. Put

$$Q_i := q_{j_i}, \qquad r_i := j_{i+1} - j_i, \qquad \mu_i := \frac{Q_{i+1}}{Q_i}. \tag{4}$$

For a gap word $w = (c_0, \ldots, c_{r-1})$, write

$$B_0(w) := 0, \qquad B_h(w) := \sum_{u<h} c_u, \qquad C(w) := \sum_{h=0}^{r-1} 3^{r-1-h} 2^{B_h(w)}. \tag{5}$$

Theorem 1 (record-renewal equation)**.** *For every consecutive tail-record gap,*

$$\boxed{e_{j_i} = \frac{C(w_i) + 2^{B_{r_i}(w_i)} e_{j_{i+1}}}{3^{r_i}}.} \tag{6}$$

Equivalently, if

$$\mathcal{R}_i := \sum_{h=0}^{r_i-1} \frac{Q_i}{q_{j_i+h}}, \tag{7}$$

then

$$\boxed{\mathcal{R}_i = \frac{C(w_i)}{3^{r_i-1}}, \qquad e_{j_i} = \frac{\mathcal{R}_i}{3} + \frac{e_{j_{i+1}}}{\mu_i}.} \tag{8}$$

Proof. Split the series in (3) at j_{i+1}. The part inside the gap gives $\mathcal{R}_i/3$, while the remaining tail is

$$\frac{Q_i}{3} \sum_{h=j_{i+1}}^{\infty} \frac{1}{q_h} = \frac{Q_i}{Q_{i+1}} e_{j_{i+1}} = \frac{e_{j_{i+1}}}{\mu_i}.$$

Moreover,

$$\mathcal{R}_i = \sum_{h=0}^{r_i-1} \frac{2^{B_h(w_i)}}{3^h} = \frac{C(w_i)}{3^{r_i-1}}.$$

This proves (8). Since $\mu_i = 3^{r_i}/2^{B_{r_i}(w_i)}$, equation (6) follows. □

Theorem 2 (four-map companion confinement)**.** *Suppose that, from some record onward, every gap lies inside its deficit-four horizon. Relabel that tail so that this holds for every $i \geq 0$. Then every gap word belongs to*

$$\mathcal{A}_4 := \{1, 12, 1113, 1122\}. \tag{9}$$

At record times, the companion obeys the backward iterated system

$$e_{j_i} = F_{w_i}(e_{j_{i+1}}), \tag{10}$$

where

w	$C(w)$	$\mu(w)$	$F_w(u)$
1	1	3/2	$(2u+1)/3$
12	5	9/8	$(8u+5)/9$
1113	65	81/64	$(64u+65)/81$
1122	73	81/64	$(64u+73)/81.$

(11)

Consequently,

$$\boxed{1 < e_{j_i} \leq 5 \qquad (i \geq 0).} \tag{12}$$

At every state of the corresponding raw accelerated tail,

$$\boxed{1 < e_t \leq \frac{41}{3}, \qquad 2 \leq \lceil e_t \rceil \leq 14.} \tag{13}$$

Proof. The interior record-renewal theorem gives (9). Substitution of the four words into (5)–(6) gives the table (11). Each map in (11) sends the closed interval $[1,5]$ into itself. Its slope is $1/\mu(w)$, and

$$\frac{1}{\mu(w)} \le \frac{8}{9} \qquad (w \in \mathcal{A}_4). \tag{14}$$

Fix $i < N$ and put

$$\Phi_{i,N} := F_{w_i} \circ F_{w_{i+1}} \circ \cdots \circ F_{w_{N-1}}.$$

Equation (10) gives

$$e_{j_i} = \Phi_{i,N}(e_{j_N}). \tag{15}$$

The slope of $\Phi_{i,N}$ is exactly Q_i/Q_N. Hence

$$|e_{j_i} - \Phi_{i,N}(1)| = \frac{Q_i}{Q_N}|e_{j_N} - 1| \longrightarrow 0, \tag{16}$$

because $e_{j_N}/Q_N \to 0$ and $Q_N \to \infty$. On the other hand, $\Phi_{i,N}(1) \in [1,5]$ for every N. Letting $N \to \infty$ proves $e_{j_i} \in [1,5]$. The strict companion floor gives $e_{j_i} > 1$, which proves (12).

It remains to bound the states inside one atom. Starting with an endpoint bound $e_{j_{i+1}} \le 5$ and applying

$$e_t = \frac{1 + 2^{a_t} e_{t+1}}{3} \tag{17}$$

backward through each word gives these upper bounds at successive positions:

w	$(e_{j_i}, e_{j_i+1}, \ldots, e_{j_{i+1}})$
1	$(11/3, 5)$
12	$(5, 7, 5)$
1113	$(385/81, 179/27, 85/9, 41/3, 5)$
1122	$(131/27, 61/9, 29/3, 7, 5)$.

(18)

The largest entry is $41/3$. Together with the strict companion floor, this proves (13). □

Put

$$m_t := \lceil e_t \rceil, \qquad \mathcal{L}_t := \log \frac{q_t}{m_t}. \tag{19}$$

Corollary 3 (linear clock drift on an interior-renewal tail)**.** *Under the hypotheses of Theorem 2,*

$$\boxed{\mathcal{L}_{j_i} - \mathcal{L}_{j_0} \geq i \log \frac{9}{8} - \log \frac{5}{2}.} \tag{20}$$

In particular, the global logarithmic separation clock has positive lower drift per coefficient record:

$$\liminf_{i\to\infty} \frac{\mathcal{L}_{j_i}}{i} \geq \log \frac{9}{8}. \tag{21}$$

The ceiling defects on this tail belong to the finite set

$$d_t := 2^{a_t} m_{t+1} - 3m_t + 1 \in \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}. \tag{22}$$

Proof. Every multiplier in (11) is at least $9/8$, so

$$\frac{Q_i}{Q_0} \geq \left(\frac{9}{8}\right)^i.$$

At record times, (12) gives $2 \leq m_{j_i} \leq 5$. Therefore

$$\begin{aligned} \mathcal{L}_{j_i} - \mathcal{L}_{j_0} &= \log \frac{Q_i}{Q_0} - \log \frac{m_{j_i}}{m_{j_0}} \\ &\geq i \log \frac{9}{8} - \log \frac{5}{2}, \end{aligned}$$

which proves (20)–(21). The words in (9) use only exponents $a_t \leq 3$. The general ceiling-defect bound $-2 \leq d_t \leq 2^{a_t} - 1$ then gives (22). □

Corollary 4 (horizon-exit versus bounded-companion dichotomy)**.** *Every normalized hypothetical divergent orbit satisfies at least one of the following alternatives.*

1. *It has infinitely many deficit-four horizon exits at consecutive coefficient records.*
2. *On a tail, its synchronized companion satisfies (12)–(13), its ceiling and defect alphabets are finite as in (13) and (22), and its logarithmic clock satisfies the linear drift (20).*

Proof. If horizon exits are finite, every sufficiently late consecutive record gap lies inside its current horizon. Apply Theorem 2 and Corollary 3 after discarding the finite prefix. □